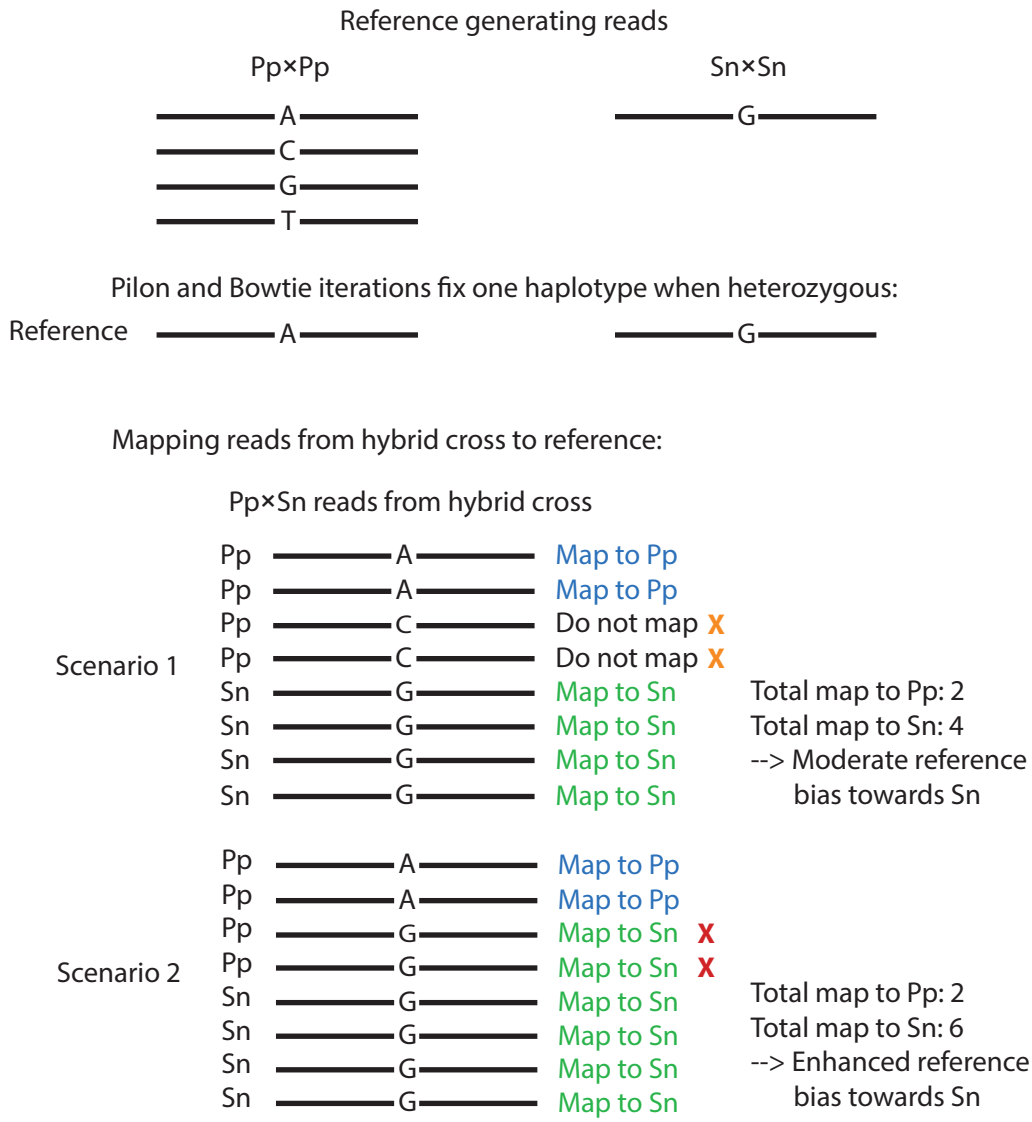


**Supplementary Figure S4. Mapping behavior of sequencing reads derived from heterozygous populations mapped to a fixed reference transcriptome.** Intrapopulation crosses were sequenced to generate reads to make two new population specific references, designated as reference Pp×Pp and reference Sn×Sn. At a particular single nucleotide polymorphism (SNP) position, reference Pp×Pp is postulated to be heterozygous with two-four possible haplotypes (A, C, G, or T). However, due to the fixation of only one haplotype during Pilon-Bowtie iterations, the reference transcriptome for Pp×Pp used in the final analysis represents in this example, the allele 'A' at this SNP position. Conversely, Reference Sn×Sn is homozygous at the same SNP position, consistently showing the allele 'G'. The sequencing reads utilized in this experiment are derived from a hybrid of the two references, containing a mix of alleles from both population Pp and population Sn. In this example two separate scenarios are presented that will lead to a map bias towards the Sn×Sn reference to a varying degree, while the expected map ratio should be 50/50 in the hybrid.